



FORMAL ASSESSMENT

NATURAL SCIENCES

Grade 7 Term 4 Task 1

Controlled Test (01)



Name: _____
Date: _____
Educator: _____
Moderator: _____

Class: _____
Time: _____ minutes

80

%

7

Instructions to the learners:

1. This question paper has three Sections:

Section A: Concepts [_____/20]

Section B: Energy and change [_____/36]

Section C: Planet Earth and Beyond [_____/24]

2. Read all the questions carefully.
3. Answer all the questions.
4. Write clearly and legibly with a blue pen.
5. Pay attention to spelling and sentence construction.
6. Read the given text carefully to help you answer the questions.
7. This paper consists of 13 pages.



Question 1:

- 1.1. Choose the correct answer by only circling the letter e.g. 1.7 **A.**
- 1.1.1. Which of the following is an example of wasted energy in a light bulb? (1)
- A. Light
 - B. Heat
 - C. Electricity
 - D. Sound
- 1.1.2. Which of the following is a NON-RENEWABLE energy source? (1)
- A. Solar power
 - B. Wind power
 - C. Coal
 - D. Hydropower
- 1.1.3. How does burning fossil fuels, like coal and oil, contribute to environmental problems? (1)
- A. They release greenhouse gases that trap heat in the atmosphere.
 - B. They create a limited supply of energy for the future.
 - C. They are not very efficient at producing energy.
 - D. They are difficult to transport over long distances.
- 1.1.4. A book is sitting on a shelf. What type of energy does the book have? (1)
- A. Kinetic energy
 - B. Thermal energy
 - C. Light energy
 - D. Potential energy
- 1.1.5. When a stretched rubber band is released, what happens to its potential energy? (1)
- A. It is converted to kinetic energy
 - B. It increases
 - C. It stays the same
 - D. It disappears
- 1.1.6. A battery powers a toy car. The battery loses energy as the car moves. What happens to the total amount of energy in the system (battery and car)? (1)
- A. The total energy increases.
 - B. The total energy decreases.
 - C. The total energy stays the same, but it changes form.
 - D. The energy disappears.

- 1.1.7. The Earth revolves around the Sun in a path called its: (1)
 A. Rotation
 B. Tilt
 C. Rotation
 D. Orbit
- 1.1.8. Why do we experience different seasons on Earth? (1)
 A. The Sun gets hotter and colder throughout the year.
 B. The Earth's distance to the Sun changes throughout the year.
 C. The Earth's axis is tilted as it revolves around the Sun.
 D. The Moon blocks the Sun's rays for part of the year.
- 1.1.9. What is the process called where plants capture the sun's energy? (1)
 A. Combustion
 B. Photosynthesis
 C. Respiration
 D. Transpiration
- 1.1.10. Which of the following statements is TRUE about gravity? (1)
 A. It pulls objects with a greater mass towards objects with a lesser mass.
 B. It only affects very large objects like planets and stars.
 C. It gets stronger the further apart two objects are.
 D. It only exists on Earth.

- 1.2. Match the term in column A with the definition in Column B. Write the answer below. (5)

Column A	Column B
1.2.1. Conduction	A. A current in a fluid that results from convection.
1.2.2. Convection current	B. An expert in the field of geology, the study of what the earth is made of and how it was formed.
1.2.3. Solar power	C. A person who is trained to travel in a spacecraft.
1.2.4. Geologist	D. Power obtained by harnessing the energy of the sun's rays.
1.2.5. Astronaut	E. The process of which heat, or electricity passes through a material.

1.2.1.	1.2.2.	1.2.3.	1.2.4.	1.2.5.
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1.3. Give one word for the following definition: (5)

1.3.1. Power obtained by harnessing the energy produced by waves at sea.	
1.3.2. Potential energy that is stored when a body is deformed.	
1.3.3. Energy produced by heat	
1.3.4. An imaginary line around which a body rotates.	
1.3.5. A series of organisms each dependent on the next as a source of food.	

Section B

Question 2:

Answer the following questions.

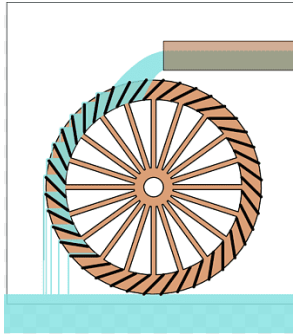
2.1. Define kinetic energy. (1)

2.2. Look at the picture of the two vehicles.



2.2.1. Why does the truck at the top of the hill have more potential energy than a car speeding down the hill? (2)

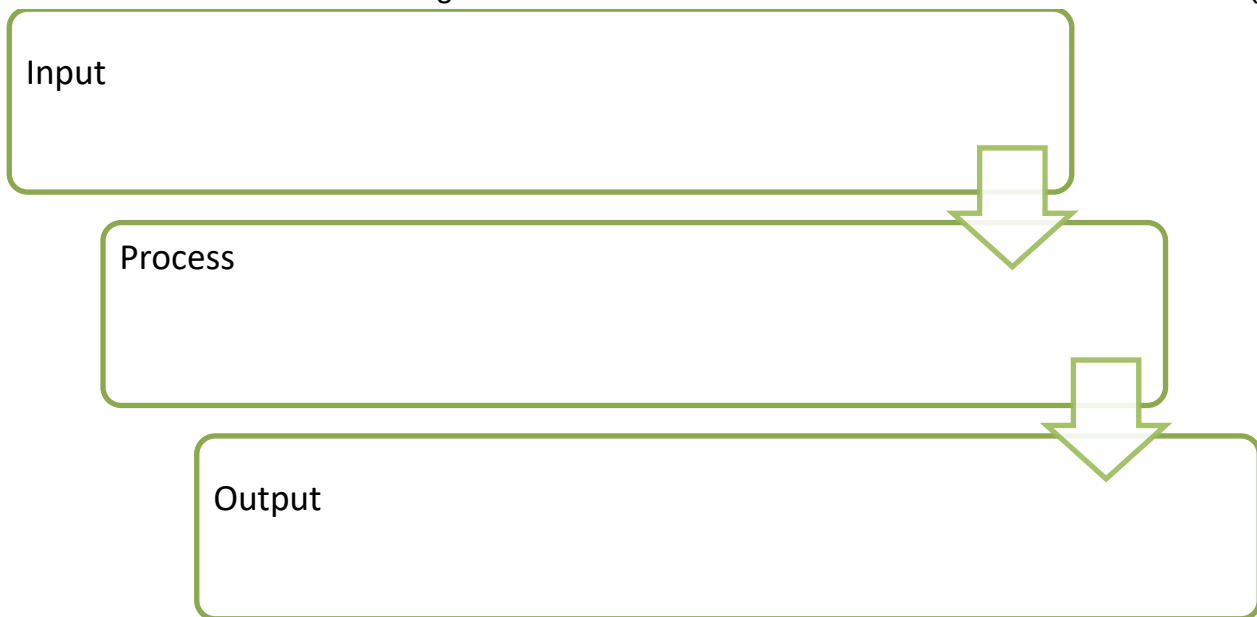
2.3. Look at the picture of a water wheel.



[PNGWING, *Waterwheel*, 25 June 2024, < <https://www.pngwing.com/en/search?q=water+Wheel>>]

2.3.1. Fill in the flow diagram.

(3)



2.4. Read the case study and answer the questions that follow.

Water Reservoirs

Water reservoirs are built to provide water for towns. They are built uphill from the houses in a town. In this way, the stored water has enough potential energy to flow when a tap in the town is opened. The height of the reservoir determines how strongly the water will flow from the taps.

When the water reservoir is not much higher than the houses, the water only trickles out of the taps. Water in reservoirs that are positioned lower than the houses, or water from boreholes and wells, must be pumped up to the houses. This is because the water below the level of the houses does not have enough energy to reach the houses.



[Laerskool Van Dyk Primary, 27 June 2024,
<https://drive.google.com/file/d/1MroBwwAr_uByiIKWfBrCgjsXh-iqu3py/view>]

2.4.1. What are the three types of potential energy? (3)

2.4.2. What type of potential energy does a water reservoir have? (1)

[10]

Question 3:

Answer the questions below.

3.1. Explain the principles of convection and heat energy in the following situations. (6)

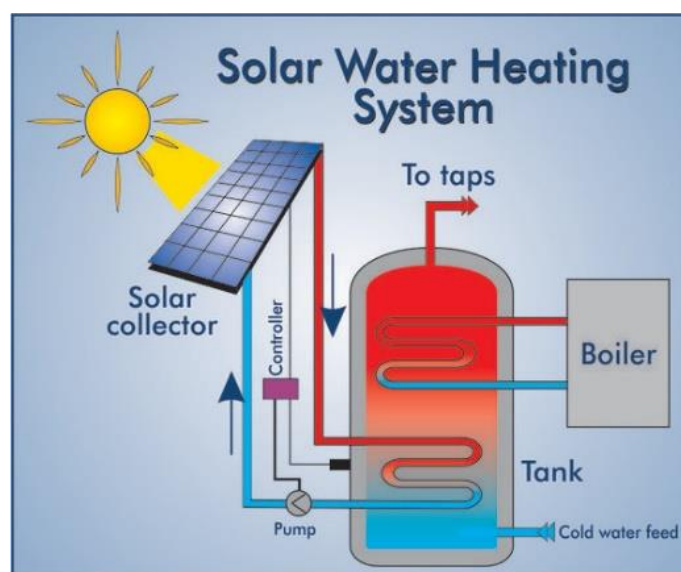
- 3.2. You're building a doghouse in a hot climate and want to keep it cool in the summer, how would you choose a roof colour to minimize heat gain inside the doghouse. Give an explanation. (2)



- 3.3. There are three spoons (plastic, metal and wooden spoon) in the bowl. Which spoon would be the best conductor of heat? Give a detailed explanation. (2)

[10]

Question 4:



[Pinterest, 27 June 2024, <<https://za.pinterest.com/pin/415668240579238791/>>]

4.1. Why do you think the pipes and tank in a solar water heating system are insulated.

(1)

4.2. You have bought a second-hand solar water system at a very good price. After a few days, you notice that the system is heating up the water but not keeping it hot. What could the problem be? Give a detailed explanation.

(2)

4.3. Discuss how a solar water heating system would work, after analysing the above image.

(5)

4.4. Based on the solar water heating system, explain how this system can save energy compared to traditional water heating methods using electricity.

(2)

[10]

Question 5:

Answer the following questions.

5.1. Underline two examples of useful energy. (2)

- a. Light bulb illuminating a room
- b. Smoke rising from a chimney
- c. Food giving you energy to run
- d. Leaving a light on in an empty room

5.2. Identify one way you can conserve energy at home. (1)

5.3. Does electricity get wasted when an appliance is on standby mode? Give a reason for your answer. (1)

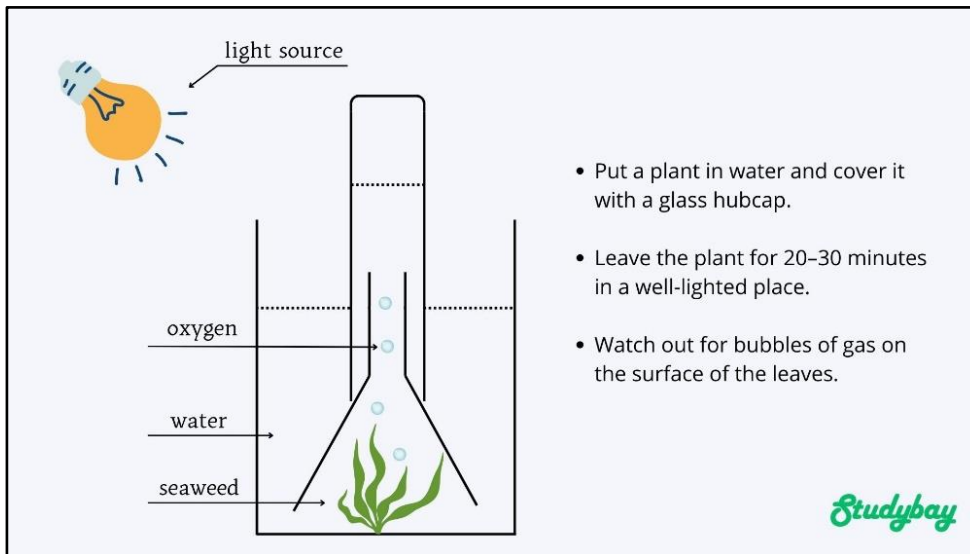
5.4. You notice your phone gets warm while charging. Is this useful or wasted energy or both? Give an explanation. (2)

[6]

Section C:**Question 6:**

Answer the questions below.

6.1. Read the description of the experiment and the sketch.



[Studybay, 27 June 2024, < <https://studybay.com/blog/photosynthesis-lab-report/>>]

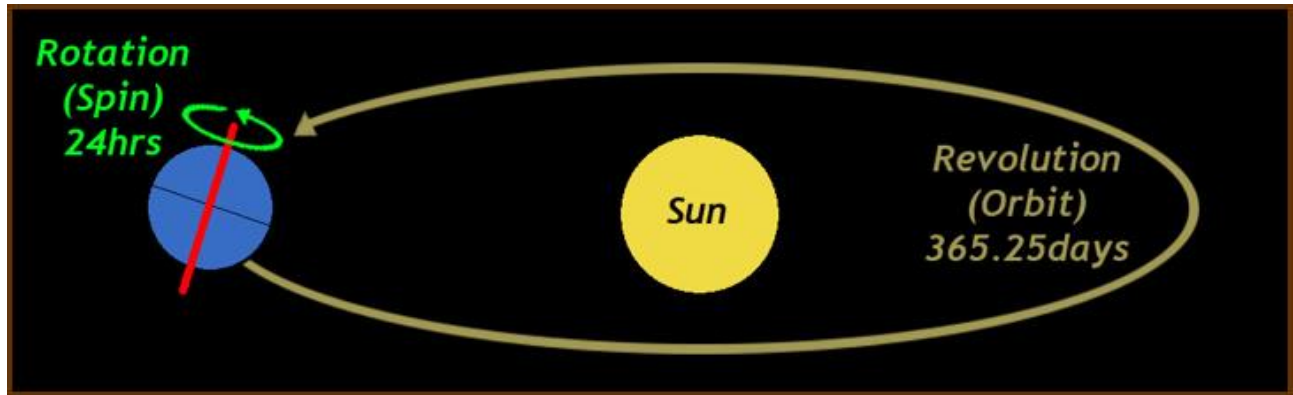
6.1.1. What is the process by which plants produce food? (1)

6.1.2. What gas does the plant take in during this process? (1)

6.1.3. In the image, what is the purpose of placing the plant in a container of water and covering it? (1)

6.1.4. Based on your knowledge of photosynthesis, what are the raw materials that a plant needs to carry out this process? (3)

6.2. Look at the diagram and answer the questions below.



[Weebly, 27 June 2024, < <https://planetearth1p.weebly.com/earths-revolution.html>>]

- 6.2.1. Earth's orbit around the Sun is not perfectly circular, but slightly elliptical. How might this slight difference in Earth's distance to the Sun throughout the year influence seasonal temperature? (3)

- 6.2.2. Why do the seasons differ in the different hemispheres? (2)

6.3. Where does the energy in coal originally come from? Give a detailed explanation. (2)

6.4. Fossil fuels are non-renewable resources. Give one positive and one negative to changing to a renewable energy source. (2)

Positive	
Negative	

Question 7:

7.1. How many phases of the Moon are there? (1)

7.2. What is the phase of the Moon called after First Quarter? (1)

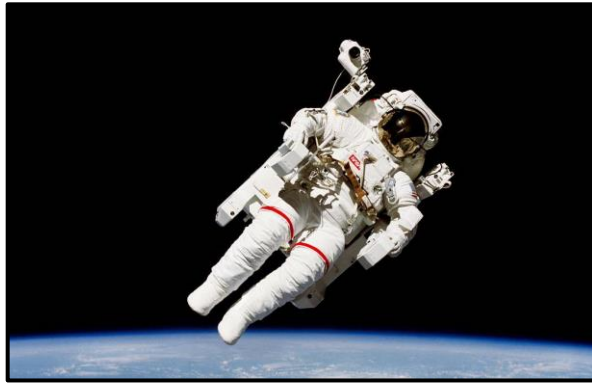
7.3. Explain the First Quarter phase of the Moon according to the criteria.

7.3.1. What can be seen? (1)

7.3.2. What happens? (1)

7.4. What has the most mass in our Solar System? (1)

7.5. Look at the picture of the astronaut below and answer the questions.



[Astronaut in space, NASA, 27 June 2024,
<https://drive.google.com/file/d/1MroBwwAr_uByilKWfBrCgjsXh-iqa3py/view>]

<https://www.nasa.gov/image-article/astronaut-bruce-mccandless-floating-free/>

7.5.1. An astronaut in space floats. Does this mean there is no gravity in space?
Explain. (2)

7.5.2. How would the weight of an object on the Moon compare to its weight on
Earth? Give a detailed explanation. (2)

[10]

TOTAL: 80